Experiment Name : 8-bit Parallel Adder Circuit Design and Observation

1. Objective

To design and simulate an 8-bit Parallel Adder Circuit using IC 74LS83.

2. Apparatus / Tools

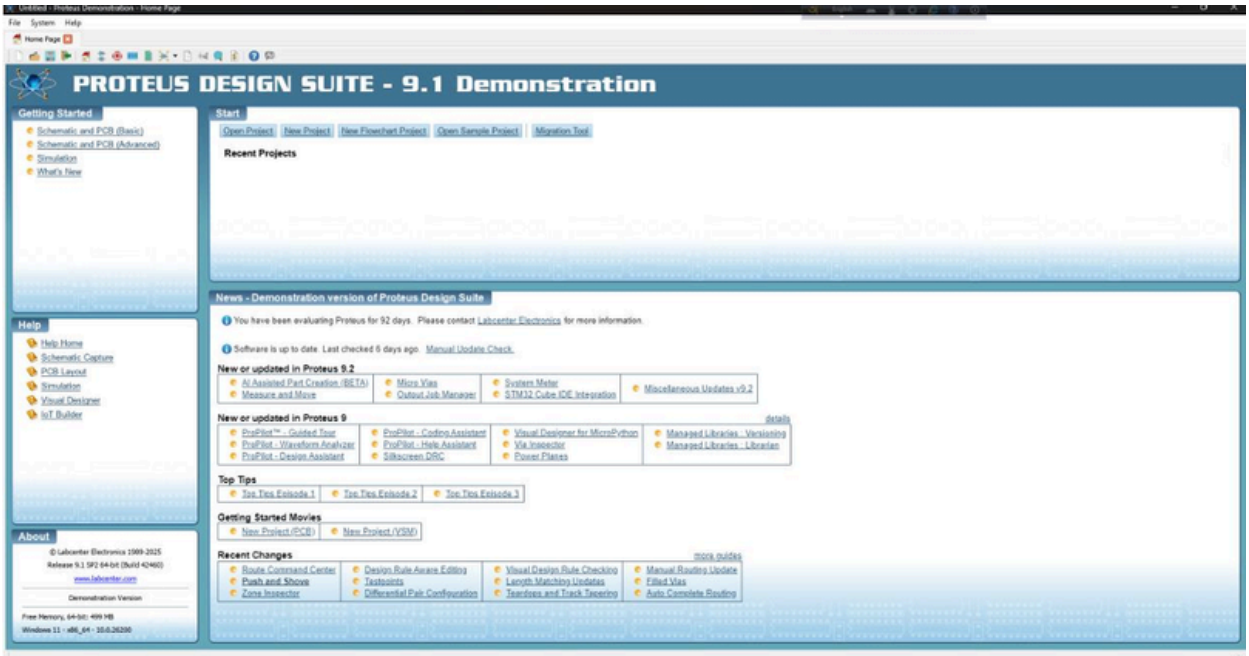
- Proteus Simulation Software
- IC 74LS83 (4-bit Binary Adder) – 2 Pieces
- Logic Toggles (For Input)
- Logic Probe / Logic State / LEDs (For Output)

3. Theory:

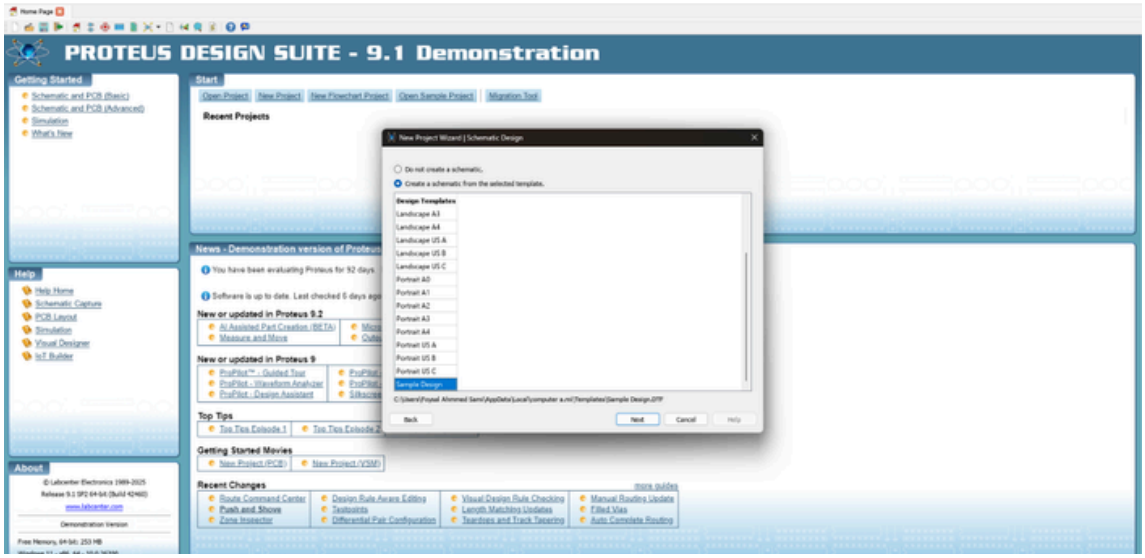
An 8-bit Parallel Adder is implemented using two IC 74LS83. To perform 8-bit addition, the two ICs are connected in cascade. The carry output (C4) of the first IC is connected to the carry input (C0) of the second IC.

4. Circuit Connection / Working Procedure

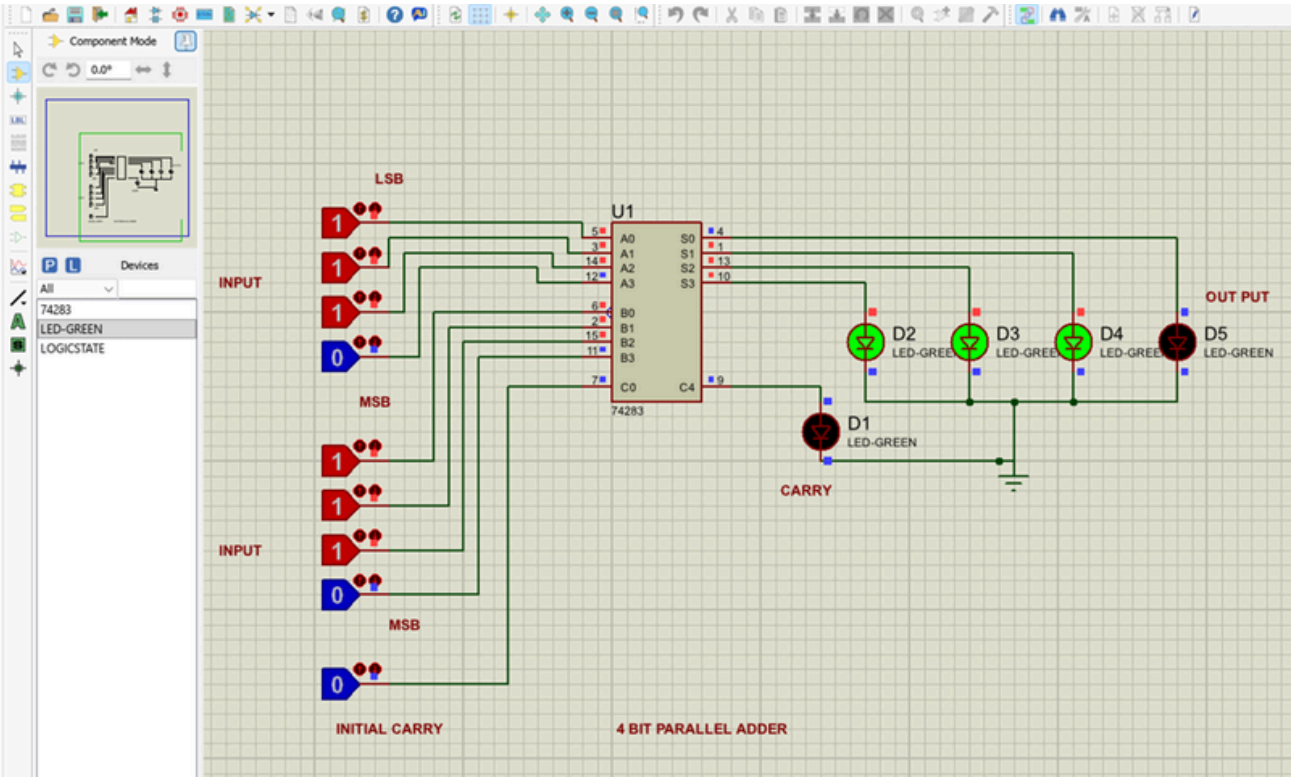
Step 1: Open the Proteus Design Suite software and click on the New Project button from the Start section on the Home Page.



Step 2 : Select the "Create a schematic from the selected template" option, choose Sample Design (or DEFAULT) from the Design Templates list, and click Next to open the workspace.



Step 3 : Pick the required components (74283, LOGICSTATE, LED-GREEN, GROUND) using Component Mode (P), complete the circuit wiring with inputs and outputs, and press the Play button to run the simulation.



Conclusion: The 8-bit Parallel Adder was successfully designed and simulated in Proteus, accurately performing binary addition using two cascaded ICs and displaying the correct sum and carry on the LEDs.

- Md. Mahedi Hasan